

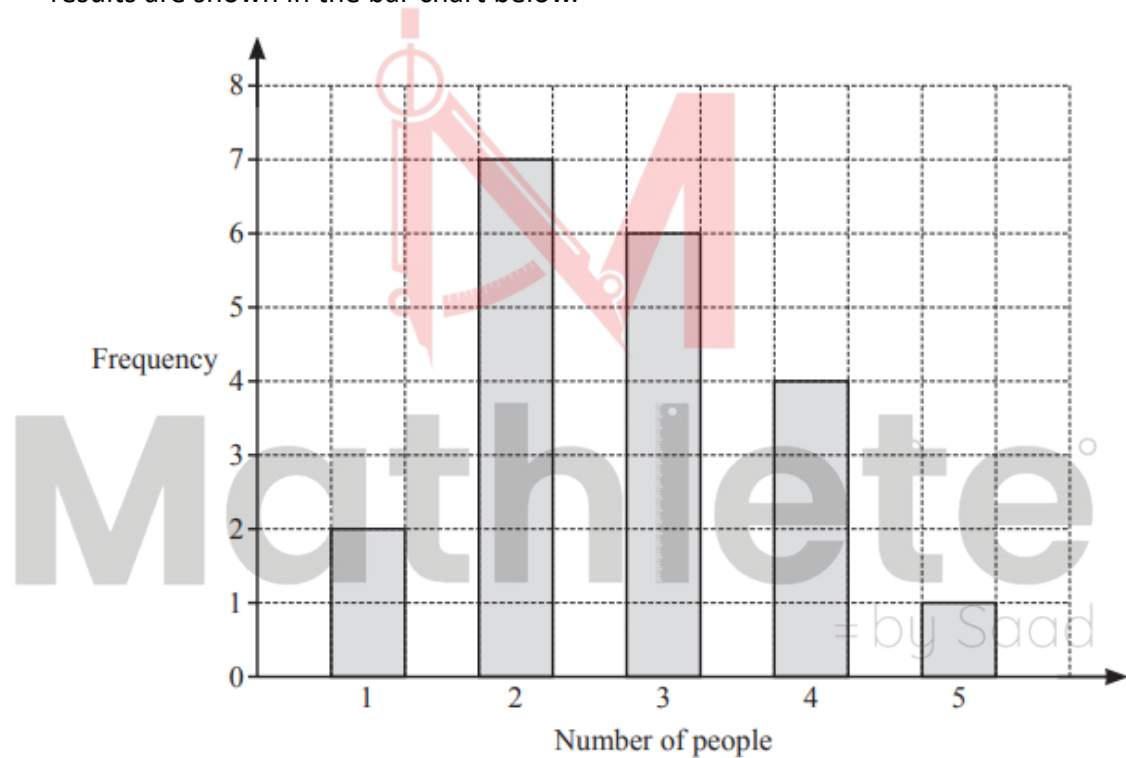
Statistics

Worksheet 2a (Non-Calculator)

Bar Chart, Pie chart and Pictogram

1. S20/qp11/q12

Aadil observed the number of people in each of 20 cars entering a car park. The results are shown in the bar chart below.



- Write down the mode. [1]
- Calculate the mean number of people in each car. [2]

2. S19/qp11/q9(a)

The students in a school each choose a piece of fruit to eat with their lunch. They can choose from either an apple, a banana or an orange.

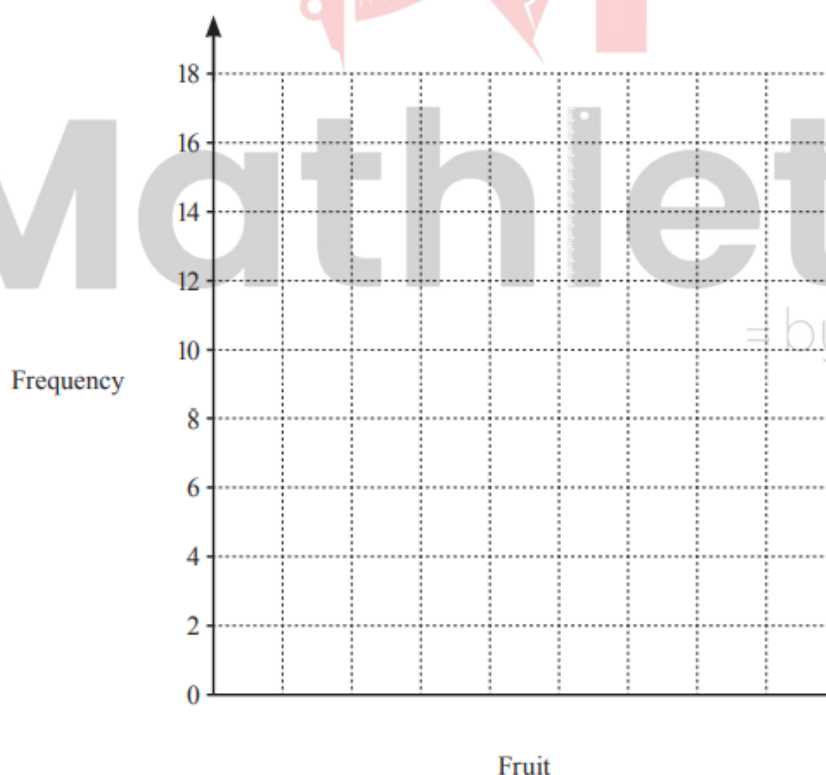
On Monday, Klaudia records the fruit chosen by 30 of her classmates. Their choices are given below.

Orange	Apple	Orange	Apple	Apple	Orange
Banana	Orange	Apple	Banana	Orange	Orange
Apple	Banana	Orange	Apple	Orange	Banana
Orange	Apple	Banana	Apple	Banana	Orange
Apple	Orange	Orange	Banana	Apple	Banana

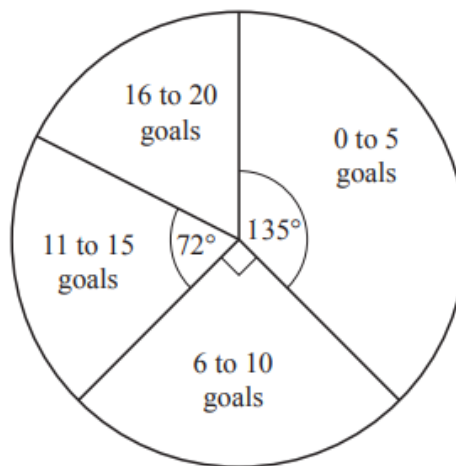
- i. Complete the frequency table for the data. [1]

Fruit	Apple	Banana	Orange
Frequency			

- ii. On the grid below, draw a bar chart to show the data. [2]



3. S20/qp12/q6



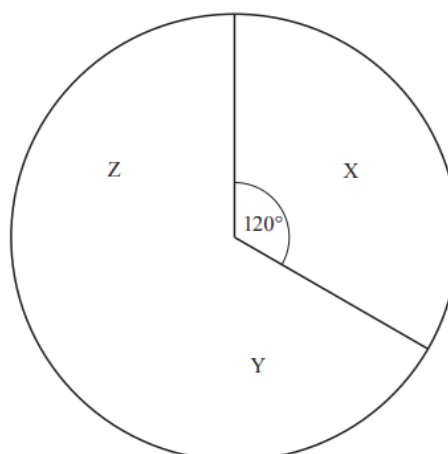
The pie chart shows information about the number of goals scored by each player in a football club.

- Write down the modal class. [1]
- 8 of the players each scored 11 to 15 goals. Work out the total number of players in the club. [2]

4. W17/qp11/q13

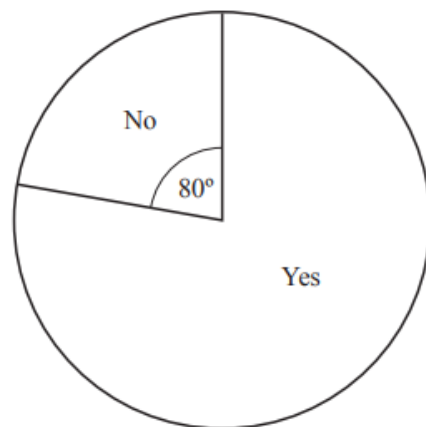
In a survey, some people were asked which of three types of tea, labelled X, Y and Z, they preferred. The diagram shows part of a pie chart that illustrates the results. The angle of the sector that represents the people who preferred Y is 168° .

- Complete the pie chart. [1]



- Find the fraction of people who preferred Y. Express your answer in its simplest form. [1]
- Given that 80 people preferred X, calculate the number of people in the survey. [1]

5. S17/qp12/q4



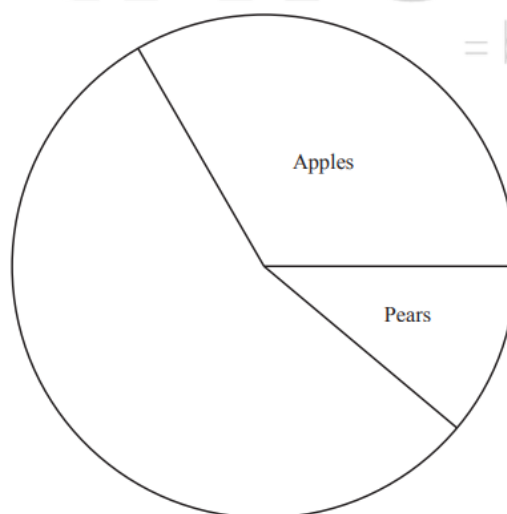
A group of students were asked if they wanted a later start to the school day. The pie chart summarises the results. 200 students said no. Work out the number of students who said yes. [2]

6. W15/qp11/q21

The table shows the masses of different fruits sold at a market stall on one day.

Fruit	Apples	Pears	Oranges	Bananas	Total
Mass (kg)	30	10	18	32	90

- Complete the pie chart to illustrate the data. [2]
- The stallholder buys apples for 60 cents per kilogram. She sells them all for 72 cents per kilogram. Calculate her percentage profit. [2]



7. W13/qp12/q4 (0580)

The pictogram shows information about the numbers of different drinks sold in a café in one hour.

Coffee	     
Tea	   
Hot chocolate	
Juice	  

Key:  represents 4 cups

- In this hour, 14 cups of hot chocolate were sold. Complete the pictogram using this information. [1]
- How many more cups of coffee than cups of tea were sold? [1]

8. W19/qp22/q6 (0580)

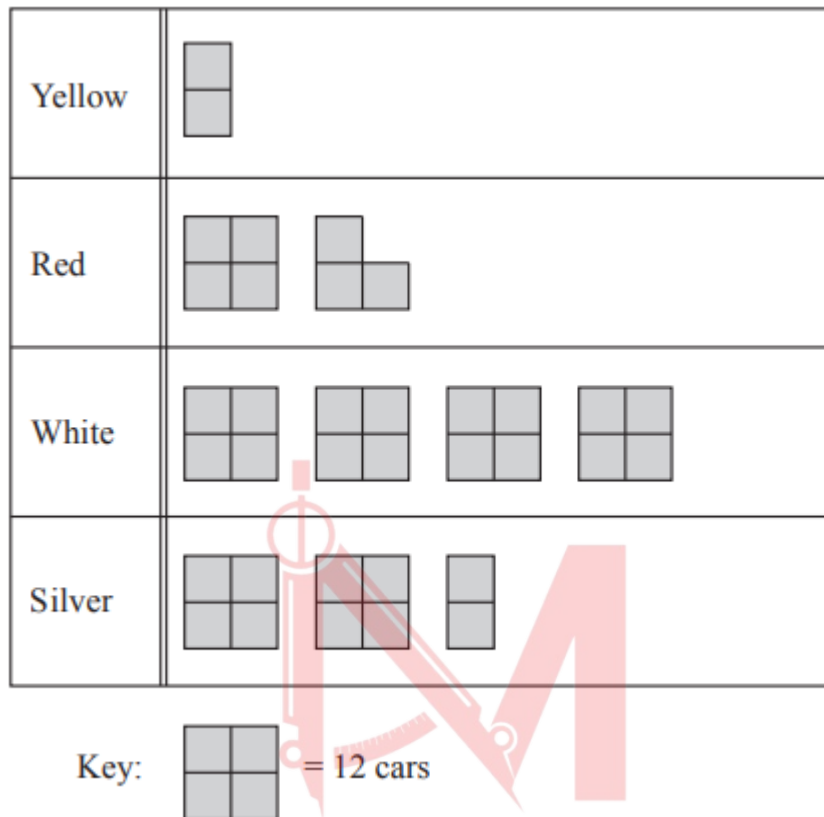
The table shows how children in Ivan's class travel to school.

Travel to school	Number of children
Walk	12
Car	7
Bicycle	9
Bus	4

Ivan wants to draw a pie chart to show this information. Find the sector angle for children who walk to school. [2]

9. M22/qp12/q4 (0580)

The pictogram shows the number of different coloured cars a garage sells in a month.



Work out how many more white cars than red cars the garage sells. [1]

Mathlete[®]
= by Saad

Answer Key

1. S20/qp11/q12

12(a)	2	1	
12(b)	2.75	2	M1 for $\frac{2 \times 1 + 7 \times 2 + 6 \times 3 + 4 \times 4 + 5 \times 1}{20}$ oe

2. S19/qp11/q9(a)

9(a)(i)	Apple 10 Banana 8 Orange 12	1	
9(a)(ii)	Correct bar chart	2	FT <i>their</i> frequency table B1 for bars labelled correctly and of equal width or bars drawn to correct height.

3. S20/qp12/q6

6(a)	0 to 5	1	
6(b)	40	2	M1 for $[8 \times] \frac{360}{72}$ or $\frac{72}{8}$

4. W17/qp11/q13

13(a)	Correct line	1	
13(b)	$\frac{7}{15}$ cao	1	
13(c)	240	1	

5. S17/qp12/q4

4	700	2	C1 for answer 900 or M1 for $\frac{200}{80} \times (360 - 80)$ oe
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6. W15/qp11/q21

21 (a)	Pie chart completed accurately, and labelled with Bananas and Oranges	2 *	M1 for $4 \times 18 (= 72)$ oe or for $4 \times 32 (= 128)$ oe
(b)	20	2 *	M1 for $\frac{72 - 60}{60} \times 100$ oe

7. W13/qp12/q4 (0580)

4	(a) 3.5 symbols in hot chocolate row	1	
	(b) 7	1	

8. W19/qp22/q6 (0580)

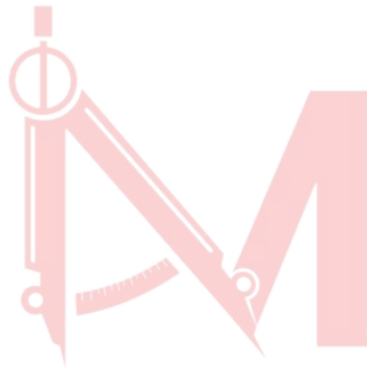
$$6 \quad \left| \quad 135 \right.$$

$$2 \quad \left| \quad \begin{array}{l} \mathbf{M1} \text{ for } \frac{12}{12+7+9+4} [\times 360] \\ \text{or } \frac{360}{12+7+9+4} [\times 12] \quad \text{oe} \end{array} \right.$$

9. M22/qp12/q4 (0580)

$$4 \quad \left| \quad 27 \right.$$

$$\left| \quad \mathbf{1} \right|$$



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